

Final Project

PSTAT122: Design and Analysis of Experiments

Spring 2026

 Due Date

Due Date: Monday, June 8, 2026, 11:59 PM

1 Overview

As a culmination of the course, students will work in groups of 3–4 members **from the same section** to design, conduct, and analyze a small-scale experiment applying the concepts learned throughout the quarter. You are expected to:

- Propose a well-structured experiment using proper design principles.
- Carry out the experiment (in a practical, low-cost, short-duration setting).
- Analyze the data using appropriate statistical tools in R.
- Submit a concise and professional written report.

2 Project Goals

- Apply key principles: randomization, replication, and blocking.
- Develop and justify treatment structures.
- Choose an appropriate design (e.g., CRD, RCBD, factorial, etc.).
- Demonstrate the full cycle: design → data → analysis → interpretation.
- Practice clear communication of experimental work.

3 Report Requirements

Your report should be a structured, professional document (in .pdf format) (approx. 5–7 pages, excluding code and appendix) should include:

3.1 Introduction

- Clear statement of the objective or research question.
- Brief context or motivation.

3.2 Experimental Design

- Description of factors and treatment structure.
- Clearly state what you are measuring and the units. Examples: Number of words recalled (count), reaction time (seconds), taste rating (1–5 scale).
- Identify which factors are fixed vs. random.
- Description of design type (CRD, RCBD, factorial, etc.).
- Explain how randomization, replication, and (if used) blocking were implemented.
- Sample size: Provide number of observations per condition. Guideline: 5–10 per treatment for CRD, 3–5 blocks for RCBD, total feasible within 1 hour.

(You are encouraged to explore more resources for determining the sample size)

3.3 Data Collection

- **Procedure:** Describe how and when the experiment was conducted (e.g., location, date, steps taken).
- **Challenges/Adjustments:** Mention any difficulties or changes made during data collection (e.g., technical issues, time adjustments).
- **Data Presentation:** Display the collected data in tables or graphs, summarizing key measures like mean and standard deviation..

3.4 Analysis

- **Exploratory Data:** Start with basic statistics (mean, SD) and visualizations (e.g., boxplots) to understand the data.
- **Hypothesis Testing:** Test your hypothesis with an appropriate statistical test (e.g., ANOVA).
- **Tables, Figures, & Code:** Include key results (ANOVA table, post-test).
- Use R to analyze the data.
- **No R code should appear in the main text; all R code must be included in the Appendices. ##**
Conclusions
- Summarize key findings.
- Comment on limitations and possible improvements.

4 Timeline & Submission

- **Week 5:** Project guidelines distributed.
- **Final Report Due:** Monday, June 8, 2026 at 11:59 PM via Gradescope. (You can use `Final Project Template.qmd` to generate to type the final report.)

5 Grading Criteria

Component	Weight
Experimental Design & Justification	30%
Clarity of Data Collection & Execution	15%
Correctness of Analysis	25%
Quality of Writing and Visuals	20%
R Code / Reproducibility	10%

6 Tips for Students

This guidance to help you think more clearly

Concept	What to Think About
Factors & Levels	What variables are you testing? How many levels per variable?
Fixed vs. Random	Is the level of the factor of specific interest (fixed), or sampled from a larger population (random)?
Blocking	Are there sources of variability you can account for (e.g., subjects, time of day)?
Replication	How many observations per condition will you need?
Randomization	How will you assign treatments or order trials to avoid bias?
Feasibility	Can this be done in one hour or less, with easily accessible materials?

6.1 Suggested Themes

Your experiment should be:

- Simple, low-cost, and feasible to conduct within one hour.
- Designed so that meaningful variation can be observed and analyzed.

Here are some idea areas to get you started:

Category	Sample Topics
Food & Taste Tests	Does salt level affect popcorn preference? Does pan-fried vs. baked cookie taste differ?
Memory & Perception	Does music volume affect word recall? Can font style impact reading speed?
Reaction Time	Does caffeine (tea vs. water) affect reaction time using an online test?
Measurement	Compare two types of measuring tools (e.g., ruler vs. app) for object lengths.
Manufacturing	Paper plane distance with different fold designs or paper weights.

You are encouraged to be creative, just make sure your design allows for structured analysis.

6.2 Expanded Project Theme Example

- **Research Question:** Does type of background sound affect how many words a person can recall?
- **Response Variable:** Number of words recalled correctly from a list
- **Factors:** - Background sound (Fixed, 3 levels: silence, instrumental music, music with lyrics)
- **Blocking:** Could block by time of day, or subject (if each subject does all 3)
- **Design:** Either RCBD (blocking by subject) or Completely Randomized Design (CRD) if each subject does only one condition
- **Replication:** At least 5–10 subjects per condition; if using RCBD, each subject completes all 3 conditions, so total sample size = number of subjects $\times$ 3
- **Randomization:** Random assignment of condition to subjects.

 **Experiments to Avoid**

- Paper airplane flying distance experiments.
- Words recall experiments under different conditions.
- Typing speed or typing accuracy experiments under different conditions.